KCE PATHWAYS · GRADE 3 · UNIT 6: BUILDING TOGETHER · WEEK 29 · MATH

Parade Practice in New Orleans — Teacher Lesson Plan & Answer Key

Learning objectives: Students use odd and even patterns to predict sums and products (odd + odd is even; any product with an even factor is even), explain patterns in the multiplication table with the associative and commutative properties (the 4s row is double the 2s row), find patterns in the addition table, add and subtract within 1,000 (including regrouping across zeros), and solve two-step problems.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Title and name: the student types first and last name so results go to KingCretot Experience. Read the Pathway Points rules together: 2 tries per question, points only for a correct first try, and the answer and reason appear after a second miss. Introduce Mr. Donnell's young marching band getting ready for a parade in New Orleans, Louisiana, with Dion, Dina, Dominic and Divya.
2. Warm-Up: Week 28 Review (10 pts): whole numbers as fractions, fractions greater than 1 on a number line, and comparing them. If a student misses both, review "write the whole number with the same denominator first (2 = 8/4)."
3. Picture It: Parade Practice in New Orleans. Players march in rows, count their steps and check how many yards of the route are left. Jazz grew up in New Orleans, where Louis Armstrong learned to play the cornet. Ask: what is a pattern? (Something that repeats or changes by a rule.)
4. Word Wrangler: 6 vocab cards (even number, odd number, pattern, sum, product, regroup).
5. Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade3/prepost/KCE_Math_PrePost_G3W29.pdf) and completes Page 1 (Pre) before any teaching. Do not help with answers yet; it shows what they already know.
6. Odd and Even Patterns (10 pts): odd + odd is even (the two leftovers make a pair), even + even is even, odd + even is odd. Odd-or-even answers are typed as the word odd or even. Use it to catch a wrong sum without adding (45 + 36 cannot be 80).
7. Patterns in the Multiplication Table (10 pts): the 4s row is double the 2s row (associative property), the table matches across the diagonal (commutative property), and only odd × odd is odd. Tip: use a known fact and double it (3 × 8 = 24, so 6 × 8 = 48).
8. Patterns in the Addition Table (10 pts): each step right or down adds 1; the doubles on the diagonal are all even; along a slanting line, one addend goes up 1 and the other goes down 1, so the sum stays the same.
9. Adding and Subtracting Within 1,000 (10 pts): regroup across zeros (800 = 7 hundreds, 9 tens, 10 ones), add three numbers, and check subtraction by adding. Watch for taking the smaller digit from the bigger one instead of regrouping.
10. Parade Challenge: Two-Step Problems (10 pts): rows of players, beads, yards of the route and money. Ask the student to name step 1 first, then use an odd and even check.
11. Bonus Practice: The Band Room (XP only, not scored).

12. Match-Up Round (10 pts, 2 per first-try match): Set 1 matches vocabulary to meanings, Set 2 matches multiplication facts to their products, Set 3 matches addition and subtraction problems within 1,000 to their answers.
13. Fix the Sentence (10 pts, 5 each): typed answers are checked exactly (capital letters for names of people and places and the title Mr., end mark, list commas, and one verb fix); curly quotes and extra spaces are forgiven. Both serial-comma and no-serial-comma list forms are accepted. Remind the student to keep the same words in the same order.
14. Speed Arena and Keyboard Corral (XP only, not scored).
15. Boss Battle, Part 1: Dress Rehearsal (10 pts, 5 each): odd or even sums and products, regrouping across a zero (703 − 246), an addition table path, and why every product in the 6s row is even.
16. Boss Battle, Part 2: Parade Day (10 pts, 5 each): find-the-mistake items (a forgotten regrouped hundred; 9 × 7 must be odd), missing addends, a two-step beads problem, and why 5 + 8 and 6 + 7 have the same sum.
17. Pencil Break (Post): the student turns to Page 2 (Post) of the same sheet (grade3/prepost/KCE_Math_PrePost_G3W29.pdf) and completes it, plus the ★ Challenge if time allows, before seeing the score. The sheet buttons let the student open or print it again.
18. Results (Parade Ready!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows “Parade Hero!” with the congratulations message “Parade hero! You solved every problem on the parade route!” and a big confetti burst; below 85 shows “Great Effort Today!” with encouragement (“Every musician gets better with practice.”), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: WEEK 28 REVIEW (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: How many thirds are in 4 wholes? Write 4 as a fraction with a denominator of 3.	**12/3** (also accepted: 12 / 3, 12 thirds, twelve thirds)	Each whole has 3 thirds. 4 × 3 = 12, so 4 = 12/3.
w2_s0	Set 1: Which fraction is between 1 and 2 on a number line? [5/3 / 2/3 / 7/3 / 6/3]	**5/3**	1 = 3/3 and 2 = 6/3. Only 5/3 is between them. 6/3 is exactly 2, 7/3 is past 2, and 2/3 is less than 1.
w1_s1	Set 2: Compare. Type <, >, or =. 7/4? 7/6	**>** (also accepted: greater than, is greater than, greater)	The numerators are the same. Fourths are bigger pieces than sixths, so 7/4 > 7/6.
w2_s1	Set 2: 10/4 is 2 wholes and?/4 more. Type the missing numerator.	**2** (also accepted: 2 fourths, 2/4, 2 / 4, two)	2 wholes = 8/4. 10 − 8 = 2, so 10/4 is 2 wholes and 2/4 more.
w1_s2	Set 3: Type the whole number: 16/8 =?	**2** (also accepted: 2 wholes, 2 whole, two)	Count by 8s: 8, 16. That is 2 counts, so 16/8 = 2.
w2_s2	Set 3: Find the mistake. Dion says 9/4 < 2. What is wrong? [2 = 8/4, and 9 fourths is more than 8 fourths, so 9/4 > 2. / Nothing is wrong. 9/4 < 2. / 9/4 = 2, because 9 − 4 − 4 = 1. / 9/4 is less than 1, because it is a fraction.]	**2 = 8/4, and 9 fourths is more than 8 fourths, so 9/4 > 2.**	Write 2 in fourths: 2 = 8/4. Since 9 > 8, 9/4 is 1 fourth more than 2.

ODD AND EVEN PATTERNS (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
a1_s0	Set 1: Dominic adds two odd numbers. Which is true about the sum? [It is always even. / It is always odd. / It is sometimes even and sometimes odd. / It is always 10.]	**It is always even.**	Each odd number has 1 left over after making pairs. The two leftovers make one more pair, so odd + odd is always even.
a2_s0	Set 1: Is 37 + 48 odd or even? Type odd or even.	**odd** (also accepted: it is odd, it's odd, odd number, an odd number)	37 is odd and 48 is even. Odd + even is odd. Check: 37 + 48 = 85, and 85 ends in 5.
a1_s1	Set 2: 27 trumpet players and 19 drummers march together. Is the total number of players odd or even? Type odd or even.	**even** (also accepted: it is even, it's even, even number, an even number)	27 and 19 are both odd, and odd + odd is even. Check: 27 + 19 = 46, which ends in 6.
a2_s1	Set 2: Which explanation is correct? Why is odd + odd always even? [Each odd number has one left over when you make pairs. The two leftovers make a new pair, so nothing is left over. / Because odd numbers are bigger than even numbers. / Because 1 + 1 = 3. / It is not. Odd + odd is always odd.]	**Each odd number has one left over when you make pairs. The two leftovers make a new pair, so nothing is left over.**	Think of 7 + 5: 7 is 3 pairs and 1 extra, 5 is 2 pairs and 1 extra. The 2 extras make a pair: 12, which is even.
a1_s2	Set 3: Divya picks any even number and adds 7 to it. Will the sum be odd or even? Type odd or even.	**odd** (also accepted: it is odd, it's odd, odd number, an odd number)	7 is odd, and even + odd is always odd. For example, 10 + 7 = 17 and 24 + 7 = 31.
a2_s2	Set 3: Find the mistake. Dion says 45 + 36 = 80. How can you tell he is wrong without adding? [45 is odd and 36 is even, so the sum must be odd. 80 is even. The correct sum is 81. / Nothing is wrong. 45 + 36 = 80. / The sum must be even, so 80 could be right. / Two-digit numbers always add up to a number that ends in 0.]	**45 is odd and 36 is even, so the sum must be odd. 80 is even. The correct sum is 81.**	Odd + even is always odd, so an even answer like 80 cannot be right. 45 + 36 = 81.

PATTERNS IN THE MULTIPLICATION TABLE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: Is 7 × 9 odd or even? Type odd or even.	**odd** (also accepted: it is odd, it's odd, odd number, an odd number)	Both factors are odd, and odd × odd is odd. Check: 7 × 9 = 63.
b2_s0	Set 1: In the multiplication table, every number in the 4s row is... [double the number above it in the 2s row / 2 more than the number in the 2s row / half of the number in the 2s row / always an odd number]	**double the number above it in the 2s row**	4 = 2 × 2, so each 4s product is 2 times the 2s product. For example, 2 × 6 = 12 and 4 × 6 = 24.
b1_s1	Set 2: The 6s row is double the 3s row. 3 × 8 = 24. Use the pattern: 6 × 8 =?	**48**	6 = 2 × 3, so 6 × 8 is double 3 × 8. Double 24 is 48.
b2_s1	Set 2: Which product is odd? [5 × 7 / 4 × 7 / 6 × 5 / 8 × 9]	**5 × 7**	Only odd × odd makes an odd product. 5 × 7 = 35. The others each have an even factor: 28, 30, and 72.
b1_s2	Set 3: Each number in the 9s row is one group less than the number below it in the 10s row: 9 × 6 = 60 − 6 = 54. Use the pattern: 9 × 8 =?	**72**	9 = 10 − 1, so 9 × 8 is 1 group of 8 less than 10 × 8. 80 − 8 = 72.

ID	Question	Answer	Why (explain to student)
b2_s2	Set 3: Which explanation is correct? Why is the multiplication table the same on both sides of the diagonal from 1 × 1 to 10 × 10? [Because of the commutative property: 3 × 7 and 7 × 3 have the same product. / Because every product in the table is even. / Because the table only shows square numbers. / It is not the same on both sides.]	**Because of the commutative property: 3 × 7 and 7 × 3 have the same product.**	Switching the order of the factors does not change the product, so the number in row 3, column 7 matches the number in row 7, column 3.

PATTERNS IN THE ADDITION TABLE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k1_s0	Set 1: In the addition table, start at 5 + 3 = 8. Move 4 squares to the right. What is the sum in that square?	**12**	Each step to the right adds 1. 8 + 4 = 12. That square is 5 + 7 = 12.
k2_s0	Set 1: The doubles run down the diagonal: 1 + 1, 2 + 2, 3 + 3, and so on. Which is true about the doubles? [They are all even. / They are all odd. / They go up by 1 each time. / They are all equal to 10.]	**They are all even.**	A double is a number plus itself, so it always splits into 2 equal groups: 2, 4, 6, 8, and so on. They go up by 2 and are all even.
k1_s1	Set 2: Add 1 to one addend and take 1 from the other, and the sum stays the same. Fill in: 9 + 5 = 7 +?	**7**	9 + 5 = 14. Moving 2 from the 9 to the 5 gives 7 + 7 = 14.
k2_s1	Set 2: Dominic says that in the addition table, if one addend goes up by 1, the sum goes up by 1. Which example shows this? [6 + 4 = 10 and 7 + 4 = 11 / 6 + 4 = 10 and 6 + 5 = 12 / 6 + 4 = 10 and 7 + 5 = 11 / 6 + 4 = 10 and 4 + 6 = 10]	**6 + 4 = 10 and 7 + 4 = 11**	Only the first addend changed, from 6 to 7, and the sum changed from 10 to 11. The last choice shows the commutative property instead.
k1_s2	Set 3: Look at the row for 3 in an addition table that goes from 0 to 9: 3 + 0, 3 + 1, and so on up to 3 + 9. How many of those 10 sums are odd?	**5**	3 is odd. Odd + even is odd, and there are 5 even numbers from 0 to 9: 0, 2, 4, 6, 8. So 5 sums are odd: 3, 5, 7, 9, 11.
k2_s2	Set 3: Find the mistake. Divya says 8 + 5 and 5 + 8 sit in different squares of the addition table, so they must have different sums. What is wrong? [They are in different squares, but by the commutative property they have the same sum, 13. / Nothing is wrong. 8 + 5 = 13 and 5 + 8 = 14. / Addition tables never show 8 + 5. / They are in the same square.]	**They are in different squares, but by the commutative property they have the same sum, 13.**	Changing the order of the addends does not change the sum. That is why the addition table matches on both sides of its diagonal too.

ADDING AND SUBTRACTING WITHIN 1,000 (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
d1_s0	Set 1: The practice route is 700 yards long. The band has marched 284 yards. How many yards are left?	**416** (also accepted: 416 yards, 416 yd)	Regroup 700 as 6 hundreds, 9 tens, 10 ones. 10 − 4 = 6, 9 − 8 = 1, 6 − 2 = 4. So 416 yards. Check: 416 + 284 = 700.
d2_s0	Set 1: People came to watch practice: 148 on Monday, 236 on Tuesday, and 175 on Wednesday. How many people is that in all?	**559** (also accepted: 559 people)	148 + 236 = 384. Then 384 + 175 = 559.

ID	Question	Answer	Why (explain to student)
d1_s1	Set 2: The band tosses 275 strings of beads on Friday and 389 on Saturday. How many strings of beads is that in all?	**664** (also accepted: 664 strings, 664 strings of beads)	275 + 389: 5 + 9 = 14, write 4 and regroup 1 ten. 7 + 8 + 1 = 16 tens, write 6 and regroup 1 hundred. 2 + 3 + 1 = 6. So 664.
d2_s1	Set 2: Find the mistake. Dina subtracts 603 − 278 and gets 475. What went wrong? [She took the smaller digit from the bigger digit in each place instead of regrouping. 603 − 278 = 325. / Nothing is wrong. 603 − 278 = 475. / She should have added: 603 + 278 = 881. / The answer should be 335.]	**She took the smaller digit from the bigger digit in each place instead of regrouping. 603 − 278 = 325.**	You cannot take 8 ones from 3 ones, so regroup: 603 = 5 hundreds, 9 tens, 13 ones. 13 − 8 = 5, 9 − 7 = 2, 5 − 2 = 3. Check: 325 + 278 = 603.
d1_s2	Set 3: Find the missing number: 457 +? = 900	**443**	Think 900 − 457. Regroup 900 as 8 hundreds, 9 tens, 10 ones: 10 − 7 = 3, 9 − 5 = 4, 8 − 4 = 4. So 443. Check: 457 + 443 = 900.
d2_s2	Set 3: The band marched 238 steps in the first song and 467 steps in the second song. How many steps is that in all?	**705** (also accepted: 705 steps)	238 + 467: 8 + 7 = 15, write 5 and regroup. 3 + 6 + 1 = 10 tens, write 0 and regroup. 2 + 4 + 1 = 7. So 705. Even + odd is odd, and 705 is odd, so it checks.

PARADE CHALLENGE: TWO-STEP PROBLEMS (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
c1_s0	Set 1: Two steps! The band marches in 9 rows of 5 players. Then 6 players leave to rest. How many players are still marching?	**39** (also accepted: 39 players)	Step 1: 9 × 5 = 45 players. Step 2: 45 − 6 = 39 players. Check: 45 is odd and 6 is even, so the answer must be odd, and 39 is odd.
c2_s0	Set 1: Two steps! Mr. Donnell orders 450 strings of beads, then 285 more. The band tosses 598. How many strings are left?	**137** (also accepted: 137 strings)	Step 1: 450 + 285 = 735. Step 2: 735 − 598 = 137 strings are left.
c1_s1	Set 2: Two steps! There are 8 rows of 6 trombone players, plus 9 drummers. How many players is that in all?	**57** (also accepted: 57 players)	Step 1: 8 × 6 = 48 trombone players. Step 2: 48 + 9 = 57 players.
c2_s1	Set 2: Two steps! The parade route is 950 yards. The band marches 375 yards, rests, and then marches 290 more yards. How many yards are left?	**285** (also accepted: 285 yards, 285 yd)	Step 1: 375 + 290 = 665 yards marched. Step 2: 950 − 665 = 285 yards are left.
c1_s2	Set 3: Divya knows 6 × 7 = 42. 12 is double 6, so 12 × 7 is double 42. What is 12 × 7?	**84**	12 = 2 × 6, so 12 × 7 = 2 × (6 × 7) = 2 × 42 = 84.
c2_s2	Set 3: Two steps! The band raised $640 at a bake sale and $275 at a car wash. New uniforms cost $800. How many dollars are left after buying the uniforms?	**115** (also accepted: $115, 115 dollars, $115.00)	Step 1: 640 + 275 = 915 dollars. Step 2: 915 − 800 = 115 dollars are left.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match even number	**splits into pairs with none left over**	2 points if matched on the first try.

ID	Question	Answer	Why (explain to student)
match_s0_2	Set 1: Match odd number	**has 1 left over when split into pairs**	2 points if matched on the first try.
match_s0_3	Set 1: Match sum	**the answer to an addition problem**	2 points if matched on the first try.
match_s0_4	Set 1: Match product	**the answer to a multiplication problem**	2 points if matched on the first try.
match_s0_5	Set 1: Match pattern	**repeats or changes by a rule**	2 points if matched on the first try.
match_s1_1	Set 2: Match 4 × 6	**24**	2 points if matched on the first try.
match_s1_2	Set 2: Match 8 × 6	**48**	2 points if matched on the first try.
match_s1_3	Set 2: Match 3 × 9	**27**	2 points if matched on the first try.
match_s1_4	Set 2: Match 6 × 9	**54**	2 points if matched on the first try.
match_s1_5	Set 2: Match 7 × 7	**49**	2 points if matched on the first try.
match_s2_1	Set 3: Match 600 − 245	**355**	2 points if matched on the first try.
match_s2_2	Set 3: Match 348 + 276	**624**	2 points if matched on the first try.
match_s2_3	Set 3: Match 900 − 512	**388**	2 points if matched on the first try.
match_s2_4	Set 3: Match 455 + 199	**654**	2 points if matched on the first try.
match_s2_5	Set 3: Match 700 − 63	**637**	2 points if matched on the first try.

FIX THE SENTENCE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
fs1_s0	Set 1: Fix the sentence: new orleans is in louisiana	**New Orleans is in Louisiana.**	New Orleans and Louisiana are names of places, so they get capital letters. End the telling sentence with a period.
fs2_s0	Set 1: Fix the sentence: the band has trumpets drums and tubas	**The band has trumpets, drums, and tubas.** (also accepted: The band has trumpets, drums and tubas.)	Start with a capital T, put commas between the things in the list, and end with a period.
fs1_s1	Set 2: Fix the sentence: when does the parade start	**When does the parade start?**	Start with a capital W. This sentence asks something, so it ends with a question mark.
fs2_s1	Set 2: Fix the sentence: dion dina and divya play the trumpet	**Dion, Dina, and Divya play the trumpet.** (also accepted: Dion, Dina and Divya play the trumpet.)	Each name gets a capital letter, commas go between the names in the list, and the sentence ends with a period.
fs1_s2	Set 3: Fix the sentence: did mr. donnell march down canal street	**Did Mr. Donnell march down Canal Street?**	Start with a capital D. Mr. Donnell is a name and Canal Street is the name of a street, so they get capital letters. The sentence asks something, so it ends with a question mark.

ID	Question	Answer	Why (explain to student)
fs2_s2	Set 3: Fix the sentence: the drummers is ready to march	**The drummers are ready to march.**	Start with a capital T and end with a period. Drummers means more than one, so use are, not is.

BOSS BATTLE: DRESS REHEARSAL (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb1_s0	Set 1: Is 6 × 9 odd or even? Type odd or even.	**even** (also accepted: it is even, it's even, even number, an even number)	6 is even, so any product with 6 as a factor is even. Check: 6 × 9 = 54.
bb2_s0	Set 1: The route is 703 yards. The band has marched 246 yards. How many yards are left? Find 703 − 246.	**457** (also accepted: 457 yards, 457 yd)	Regroup: 703 = 6 hundreds, 9 tens, 13 ones. 13 − 6 = 7, 9 − 4 = 5, 6 − 2 = 4. So 457. Check: 457 + 246 = 703.
bb1_s1	Set 2: Which sum is odd? [35 + 48 / 35 + 47 / 48 + 26 / 19 + 19]	**35 + 48**	Odd + even is odd: 35 + 48 = 83. The others are odd + odd or even + even, so they are even: 82, 74, and 38.
bb2_s1	Set 2: The band marched 387 yards before lunch and 425 yards after lunch. How many yards is that in all?	**812** (also accepted: 812 yards, 812 yd)	387 + 425: 7 + 5 = 12, write 2 and regroup. 8 + 2 + 1 = 11 tens, write 1 and regroup. 3 + 4 + 1 = 8. So 812 yards.
bb1_s2	Set 3: Two steps! In the addition table, start at 4 + 4 = 8. Move 3 squares down and then 2 squares to the right. What is the sum in that square?	**13**	Each step down adds 1 and each step to the right adds 1. 8 + 3 = 11, and 11 + 2 = 13. That square is 7 + 6 = 13.
bb2_s2	Set 3: Which explanation is correct? Why is every product in the 6s row even? [6 = 2 × 3, so 6 times any number is 2 times something. That is a double, and doubles are even. / Because 6 is odd. / Because every product in the multiplication table is even. / Because 6 × 1 = 7.]	**6 = 2 × 3, so 6 times any number is 2 times something. That is a double, and doubles are even.**	By the associative property, 6 × 5 = 2 × (3 × 5) = 2 × 15 = 30. Doubling always makes an even number.

BOSS BATTLE: PARADE DAY (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb3_s0	Set 1: Find the mistake. Dominic says 456 + 278 = 634. He forgot to add a regrouped hundred. Type the correct sum.	**734**	6 + 8 = 14, write 4 and regroup. 5 + 7 + 1 = 13 tens, write 3 and regroup 1 hundred. 4 + 2 + 1 = 7. So 734.
bb4_s0	Set 1: Find the missing number: 285 +? = 600	**315**	Think 600 − 285. Regroup 600 as 5 hundreds, 9 tens, 10 ones: 10 − 5 = 5, 9 − 8 = 1, 5 − 2 = 3. So 315. Check: 285 + 315 = 600.
bb3_s1	Set 2: Find the mistake. Divya says 9 × 7 = 64. Without multiplying, you can tell she is wrong. Is 9 × 7 odd or even? Type odd or even.	**odd** (also accepted: it is odd, it's odd, odd number, an odd number)	Both factors are odd, so the product must be odd. 64 is even, so it cannot be right. 9 × 7 = 63.
bb4_s1	Set 2: What is 800 − 529?	**271**	Regroup 800 as 7 hundreds, 9 tens, 10 ones: 10 − 9 = 1, 9 − 2 = 7, 7 − 5 = 2. So 271. Check: 271 + 529 = 800.

ID	Question	Answer	Why (explain to student)
bb3_s2	Set 3: Two steps! The band had 520 strings of beads. They tossed 186 on the first street and 247 on the second street. How many strings are left?	**87** (also accepted: 87 strings)	Step 1: 186 + 247 = 433 tossed. Step 2: 520 − 433 = 87 strings are left.
bb4_s2	Set 3: Find the mistake. Dion says 5 + 8 and 6 + 7 must have different sums because the addends are different. What is wrong? [Both equal 13. When one addend goes up 1 and the other goes down 1, the sum stays the same. / Nothing is wrong. 5 + 8 = 13 and 6 + 7 = 14. / Both equal 12. / Sums are only the same when the addends are the same.]	**Both equal 13. When one addend goes up 1 and the other goes down 1, the sum stays the same.**	From 5 + 8 to 6 + 7, the 5 went up 1 and the 8 went down 1. They sit on the same slanting line in the addition table.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade3/prepost/KCE_Math_PrePost_G3W29.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: write 3 with a denominator of 4; is 9/4 more or less than 2?	3 = 12/4. 9/4 is more than 2, because 2 = 8/4 and 9 fourths is 1 more fourth.
2	Circle each odd number: 14, 27, 30, 41, 58, 63	27, 41 and 63 (they end in 7, 1 and 3).
3	Draw 11 dots in pairs. Is 11 odd or even?	Odd. 11 dots make 5 pairs with 1 dot left over.
4	Finish the 5s pattern: 5, 10, 15, ___, ___, ___	20, 25, 30. The ones digits take turns: 5, 0, 5, 0.
5	Add 356 + 278; show the regrouping	634. 6 + 8 = 14 (write 4, regroup 1 ten); 5 + 7 + 1 = 13 tens (write 3, regroup 1 hundred); 3 + 2 + 1 = 6. Accept any reasonable attempt before the lesson.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	Without adding, is 47 + 29 odd or even? Then find the sum	Even, because 47 and 29 are both odd and odd + odd is even. 47 + 29 = 76.
2	4 × 6 = 24; use the 8s row pattern to find 8 × 6	48. The 8s row is double the 4s row, and double 24 is 48. The associative property explains it: 8 × 6 = 2 × (4 × 6).
3	Divya says 700 − 382 = 482	She took the smaller digit from the bigger digit in the ones and tens instead of regrouping. Regroup 700 as 6 hundreds, 9 tens, 10 ones: 700 − 382 = 318. Check: 318 + 382 = 700.
4	625 strings of beads; tossed 248, then 159	Step 1: 248 + 159 = 407 tossed. Step 2: 625 − 407 = 218 strings left. (Or 625 − 248 = 377, then 377 − 159 = 218.)

★ CHALLENGE

#	Prompt	Model answer / grading note
1	4 + 9 and 5 + 8 are on the same slanting line: 3 more facts and why	Any 3 other facts with a sum of 13 that fit the table, such as 6 + 7, 7 + 6 and 8 + 5. Moving along the line, one addend goes up 1 and the other goes down 1, so the sum stays 13.